

CreditWise Loan Approval System: Intelligent Loan Approval Prediction Using Machine Learning

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Abstract—Loan approval is an important decision-making process in the banking and financial sector. Financial institutions need to evaluate different applicant characteristics, including income, credit score, savings, employment status, debt-to-income ratio, loan purpose, and other demographic and financial information before making an approval decision. Traditional manual loan assessment can be time-consuming and may introduce inconsistencies in preliminary decision-making. This paper presents CreditWise Loan Approval System, a machine learning-based approach for predicting loan approval outcomes from applicant-related data. The proposed system performs data preprocessing, exploratory data analysis, categorical encoding, feature engineering, feature scaling, model training, and performance evaluation. Three classification algorithms, namely Logistic Regression, K-Nearest Neighbors (KNN), and Gaussian Naive Bayes, were implemented and compared. The models were evaluated using accuracy, precision, recall, and F1-score. Logistic Regression and Gaussian Naive Bayes achieved an accuracy of 86.5%, whereas KNN achieved 76.0%. Logistic Regression obtained a precision of 78.33%, recall of 77.05%, and F1-score of 77.69%. Gaussian Naive Bayes achieved the highest precision of 80.36%. The experimental findings demonstrate that machine learning can provide an effective decision-support mechanism for preliminary loan approval prediction.

Index Terms—Loan Approval, Machine Learning, Logistic Regression, K-Nearest Neighbors, Gaussian Naive Bayes, Classification, Credit Score, Loan Prediction

I. INTRODUCTION

Loan approval is one of the most important activities performed by financial institutions. Banks and other lending organizations need to determine whether an applicant is financially capable of repaying a requested loan. This decision generally depends on several factors such as applicant income, coapplicant income, credit score, savings, employment status, debt-to-income ratio, marital status, education level, loan purpose, property area, and employer category.

In traditional loan processing systems, applicant information is often reviewed manually by financial officers. Although human evaluation can consider contextual information, manual processing becomes difficult when a financial institution receives a large number of applications. It may also require considerable time and resources. Furthermore, subjective judgment can introduce inconsistencies between different applications.

The development of machine learning has provided new opportunities for automated decision-support systems. Machine learning algorithms can identify patterns from historical data and use those patterns to predict outcomes for new observations. In a loan approval scenario, a model can learn the relationship between applicant characteristics and previous loan approval decisions [2], [10].

The objective of this research is to develop a machine learning-based loan approval prediction system named *CreditWise Loan Approval System*. The system analyzes applicant-related features and predicts whether a loan application is likely to be approved or rejected.

The target variable used in this research is *Loan_Aproved*. A value of 1 represents an approved loan, while a value of 0 represents a rejected loan.

Three machine learning algorithms were selected for the experimental comparison:

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Gaussian Naive Bayes

The main contributions of this study are as follows:

- Comparative evaluation using accuracy, precision, recall, and F1-score.
- Separate visualization of a Decision Tree to understand tree-based decision structures.

The remainder of this paper is organized as follows. Section II discusses related work. Section III presents the proposed methodology and machine learning models. Section IV presents the experimental results and detailed graph analysis. Section V concludes the research and discusses future directions.

II. RELATED WORK

Machine learning has been increasingly applied to financial prediction and decision-support problems. Classification algorithms are particularly useful when the target variable contains discrete outcomes such as approved and rejected.

Logistic Regression is one of the most commonly used statistical classification methods for binary classification. It estimates the probability of an outcome based on a weighted combination of input variables. Due to its relatively simple

structure and interpretable coefficients, Logistic Regression is frequently used as a baseline model [7].

K-Nearest Neighbors is another classification technique that determines the class of an unknown observation based on nearby observations in the feature space. KNN can be effective when observations belonging to the same class form distinguishable local groups. However, the performance of KNN can be affected by feature scaling and the selected value of the number of neighbors [8].

Gaussian Naive Bayes is a probabilistic classifier derived from Bayes' theorem. It assumes that the features are conditionally independent given the class. Although this assumption may not always perfectly represent real-world financial data, the algorithm is computationally efficient and can perform well on structured classification problems [9].

Machine learning approaches for loan prediction generally involve several common stages, including data preprocessing, missing value treatment, categorical encoding, feature selection, model training, and performance evaluation. The quality of preprocessing can have a significant influence on the final model performance [2], [11].

In this research, Logistic Regression, KNN, and Gaussian Naive Bayes were selected because they represent three different classification approaches: a linear probabilistic classifier, a distance-based classifier, and a probabilistic Bayesian classifier. Their performance was compared under the same experimental conditions [10], [11].

III. METHODOLOGY

The proposed CreditWise system follows a sequential machine learning workflow designed to transform raw applicant information into a reliable preliminary loan approval prediction. The complete workflow consists of Dataset, Data Cleaning, Exploratory Data Analysis, Encoding, Feature Engineering, Scaling, Train/Test Split, Model Training, Evaluation, and Prediction. This sequence provides a consistent experimental procedure in which the same general data preparation and evaluation principles are applied to the three primary classifiers.

The raw dataset contains applicant-related financial, employment, demographic, and loan-specific information. The considered attributes include income, coapplicant income, credit score, savings, employment status, debt-to-income ratio, marital status, education level, loan purpose, property area, and employer category. The target variable is *Loan_Aproved*, where a value of 1 represents an approved application and a value of 0 represents a rejected application. Such a binary target makes the problem suitable for supervised classification.

Before model training, the dataset is loaded and inspected to understand its structure, data types, missing values, and target distribution. Data cleaning is performed to handle incomplete or irrelevant information. Cleaning is important because inconsistent or missing input values can affect the learning process and can lead to unreliable predictions. The preprocessing stage therefore aims to produce a consistent

representation of applicant records before the classifiers are fitted [3], [4].

Exploratory Data Analysis (EDA) is then used to examine the distribution of the variables and the outcome classes. EDA helps identify patterns and provides an initial understanding of the characteristics of the dataset. In particular, the distribution of the target variable is important because an uneven class distribution can influence the interpretation of accuracy and other performance measures.

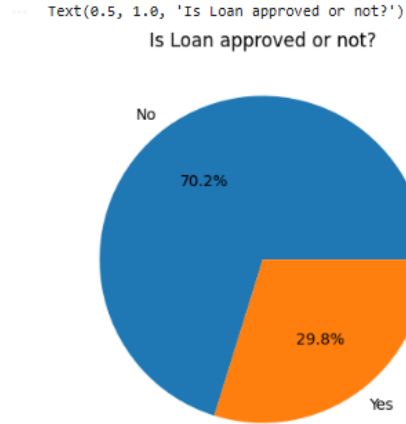


Fig. 1. Distribution of Loan Approval Outcomes

Figure 1 shows the distribution of the loan approval target. The pie chart indicates that 70.2% of the observations belong to the “No” category, while 29.8% belong to the “Yes” category. Therefore, the rejected class represents the majority of the observations, whereas the approved class represents a smaller portion of the dataset. This distribution is important for interpreting the experimental results because accuracy alone may not fully describe how effectively the classifier identifies both classes.

The relatively larger “No” portion means that a classifier can obtain a reasonable overall accuracy while still producing errors for approved applications. Consequently, precision, recall, and F1-score are considered together with accuracy. Precision indicates the reliability of positive predictions, recall indicates the ability to identify actual positive cases, and F1-score provides a balance between precision and recall. This multi-metric evaluation is particularly useful when the target classes are not evenly distributed [2], [11].

After the initial analysis, categorical variables are transformed into numerical representations so that they can be processed by machine learning algorithms. Numerical features are also examined for their ranges. Feature scaling is especially relevant to KNN because its classification decision is based on distances between observations. Without appropriate scaling, a variable with a larger numerical range may dominate the distance calculation and reduce the contribution of other relevant attributes [2], [8].

Feature engineering is applied where appropriate to obtain a useful representation of the applicant information. The

objective is not to introduce unnecessary complexity, but to provide the classifiers with informative and consistently formatted input variables. The processed feature matrix is then separated from the target variable. The resulting features are used as model inputs, while *Loan_Aproved* is retained as the prediction target.

The dataset is divided into training and testing subsets. The training data are used to learn the parameters of each classifier, while the testing data are kept separate until evaluation. This procedure allows the performance of the models to be examined on observations that were not directly used to fit the models. All three primary algorithms are trained under the same general experimental framework so that their reported results can be compared fairly.

Three classification algorithms are implemented: Logistic Regression, K-Nearest Neighbors (KNN), and Gaussian Naive Bayes. These algorithms were selected because they represent different classification strategies. Logistic Regression provides a linear probabilistic approach, KNN uses local distance relationships, and Gaussian Naive Bayes applies a probabilistic model based on Bayes' theorem. Comparing these methods makes it possible to observe how different learning assumptions behave on the same loan approval task [10], [11].

The final stage of the workflow is model evaluation and prediction. Each trained classifier produces predictions for the test observations. The predictions are compared with the actual target labels using accuracy, precision, recall, and F1-score. The resulting values are then analyzed to identify the model that provides the strongest overall balance of performance. The system is intended as a decision-support mechanism rather than a replacement for responsible financial assessment; therefore, the prediction should be interpreted as a model output based on historical patterns and available applicant information.

The workflow can also be viewed as a sequence of transformations in which each stage has a specific purpose. Data cleaning reduces inconsistencies in the raw observations, EDA provides an understanding of the target distribution, encoding converts non-numerical information into a machine-readable form, and scaling helps place numerical attributes on comparable ranges. The train/test separation then ensures that the final evaluation is performed on unseen observations. This sequential organization reduces the possibility that decisions made during preprocessing are disconnected from the final classification task.

The target distribution shown in Figure 1 also provides useful context for the model comparison. Since the "No" category is considerably larger than the "Yes" category, the evaluation should not depend on a single metric. A model that simply favors the majority class may appear effective when only accuracy is considered. However, its recall for approved applications may be comparatively weak. Reporting several complementary metrics therefore gives a more informative description of the behavior of each classifier.

The same general experimental procedure is followed for all three primary models. This is important because com-

paring algorithms under substantially different preprocessing conditions could make the comparison difficult to interpret. By maintaining a common feature preparation and evaluation framework, the observed differences are more closely associated with the characteristics of the learning algorithms themselves. The resulting comparison can therefore be used to determine which approach is most suitable for this particular dataset and prediction objective.

Finally, the predicted class can be interpreted as an indication of whether the applicant record resembles patterns associated with previous approved or rejected applications. Such a prediction should not be treated as an independent financial decision because real loan approval may require additional information, verification, policy rules, human review, and regulatory considerations. The CreditWise system is therefore positioned as a preliminary analytical and decision-support tool.

A. Logistic Regression

Logistic Regression is a supervised classification algorithm used for predicting binary outcomes.

The logistic function is defined as:

$$P(Y = 1|X) = \frac{1}{1 + e^{-z}} \quad (1)$$

where

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n \quad (2)$$

In this project, Logistic Regression predicts the probability that a loan application belongs to the approved class.

The model provides a useful baseline because it is relatively simple and computationally efficient [2], [7].

B. K-Nearest Neighbors

K-Nearest Neighbors classifies an observation based on the classes of its nearest neighbors.

The Euclidean distance between two observations can be expressed as:

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad (3)$$

In this study, KNN was trained using the processed and scaled feature space.

Scaling is particularly important for KNN because variables with larger numerical ranges can otherwise dominate the distance calculation [2], [8].

C. Gaussian Naive Bayes

Gaussian Naive Bayes uses Bayes' theorem to estimate the probability of each class.

The basic probability formulation is:

$$P(C|X) = \frac{P(X|C)P(C)}{P(X)} \quad (4)$$

For continuous numerical features, Gaussian Naive Bayes assumes that the feature values follow a Gaussian distribution within each class.

The model is computationally efficient and provides a probabilistic approach to loan approval classification [2], [9].

D. Decision Tree Visualization

A Decision Tree was also generated separately for visualization and interpretability purposes.

Unlike the three primary models used in the numerical performance comparison, the Decision Tree was not included in the final performance table because the reported experimental metrics were obtained from Logistic Regression, KNN, and Gaussian Naive Bayes.

The Decision Tree visualization illustrates how a tree-based model can divide the feature space using a sequence of conditions.

For example, a tree may first divide observations according to a financial or credit-related feature. Subsequent branches can then use additional features to reach a final classification.

This visualization is useful for explaining the concept of rule-based machine learning decisions.

IV. ANALYSIS AND RESULTS

The performance of the three classification models was evaluated using accuracy, precision, recall, and F1-score [2].

TABLE I
PERFORMANCE COMPARISON OF THE IMPLEMENTED MODELS

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	86.50%	78.33%	77.05%	77.69%
KNN	76.00%	62.75%	52.46%	57.14%
Gaussian Naive Bayes	86.50%	80.36%	73.77%	76.92%

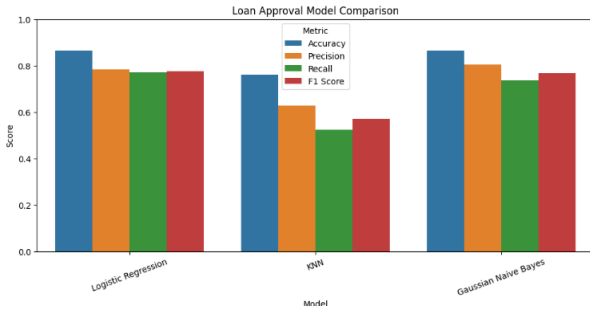


Fig. 2. Comparison of Accuracy, Precision, Recall, and F1-Score

A. Logistic Regression Results

Logistic Regression achieved an accuracy of 86.50%.

Its precision was 78.33%, meaning that a substantial proportion of applications predicted as approved were correctly classified.

The model achieved a recall of 77.05%. This indicates that Logistic Regression successfully identified a relatively high proportion of the actual approved applications.

The F1-score of 77.69% demonstrates a balanced relationship between precision and recall.

Overall, Logistic Regression provided strong and consistent performance and achieved the highest F1-score among the three models.

B. KNN Results

KNN achieved an accuracy of 76.00%.

Its precision was 62.75%, while recall was 52.46%. The resulting F1-score was 57.14%.

The relatively lower performance of KNN may be associated with the complexity of the feature space and the distance-based nature of the algorithm. Loan approval decisions can depend on multiple interacting financial and categorical characteristics, which may not always form clear local neighborhoods.

Therefore, although KNN can be useful for classification, it produced weaker results compared with Logistic Regression and Gaussian Naive Bayes in this experiment.

C. Gaussian Naive Bayes Results

Gaussian Naive Bayes achieved an accuracy of 86.50%, matching the accuracy of Logistic Regression.

The model achieved the highest precision among the three evaluated models, with a value of 80.36%.

Its recall was 73.77%, and the F1-score was 76.92%.

The results indicate that Gaussian Naive Bayes was particularly effective at maintaining a high precision level while providing competitive overall classification performance.

When the three models are considered together, Logistic Regression and Gaussian Naive Bayes clearly outperform KNN in overall accuracy. Both achieved 86.50% accuracy, while KNN achieved 76.00%. The difference suggests that the linear probabilistic representation used by Logistic Regression and the probabilistic assumptions of Gaussian Naive Bayes were more suitable for the processed feature space than the local distance relationships used by KNN. Nevertheless, accuracy alone does not determine the best model because the precision, recall, and F1-score values provide additional information about prediction quality.

Classification metrics are used to evaluate the CreditWise models. Accuracy measures the proportion of all predictions that are correct:

These metrics are considered together because the target distribution is not balanced. Recall is useful for understanding how effectively the system identifies approved applications, while precision indicates how reliable the positive predictions are. The F1-score summarizes the balance between these two measures [2].

For the CreditWise system, the distinction between false positive and false negative outcomes is important. A false positive represents an application predicted as approved even though the actual outcome is rejection. On the other hand, a false negative represents an application predicted as rejected even though the actual outcome is approval.

Therefore, confusion matrix analysis provides additional information beyond accuracy and helps understand the type of errors produced by a classification model.

The visualization is useful because a Decision Tree can be interpreted as a collection of decision rules. For example, an applicant may be directed through different branches depending on their financial and credit characteristics.

The tree was visualized separately from the primary model comparison because the main experimental performance table was specifically based on Logistic Regression, KNN, and Gaussian Naive Bayes.

Therefore, the Decision Tree figure should be interpreted as a model visualization rather than as an additional numerical result in Table I.

Overall, the experimental analysis indicates that the three models do not behave equally on the same applicant data. The lower KNN results demonstrate that a distance-based classifier may require careful tuning of neighborhood size and feature representation, while the stronger results of Logistic Regression indicate that a relatively simple model can provide a competitive baseline for this classification problem. Gaussian Naive Bayes also performs strongly and offers the advantage of a probabilistic interpretation. These observations support the use of comparative evaluation rather than relying on a single algorithm without measurement.

V. CONCLUSION AND FUTURE WORK

This research presented the CreditWise Loan Approval System, a machine learning-based approach for predicting loan approval outcomes using applicant-related financial and demographic information. The complete workflow included data preprocessing, exploratory data analysis, feature engineering, feature scaling, model training, and evaluation. Three classification algorithms were implemented: Logistic Regression, K-Nearest Neighbors, and Gaussian Naive Bayes. The experimental results showed that Logistic Regression and Gaussian Naive Bayes achieved the highest accuracy of 86.5%. Logistic Regression achieved a precision of 78.33%, recall of 77.05%, and F1-score of 77.69%. Gaussian Naive Bayes achieved a precision of 80.36%, recall of 73.77%, and F1-score of 76.92%. KNN achieved an accuracy of 76.0%, precision of 62.75%, recall of 52.46%, and F1-score of 57.14.

Based on the overall balance between precision, recall, and F1-score, Logistic Regression provided the strongest overall performance among the three evaluated models.

However, Gaussian Naive Bayes also demonstrated competitive performance and achieved the highest precision.

The Decision Tree visualization was presented separately to provide an interpretable representation of tree-based classification decisions. This visualization can be useful for understanding how different feature conditions can lead to different prediction outcomes.

- Hyperparameter optimization can be applied to improve model performance [12].
- Cross-validation can be used to obtain more robust performance estimates.

- Additional classification algorithms such as Random Forest, XGBoost, Support Vector Machine, and Gradient Boosting can be evaluated.
- ROC curves and AUC scores can be included for a more comprehensive evaluation.
- Explainable AI techniques can be incorporated to explain individual loan predictions.
- Fairness and bias analysis can be performed to identify potential discriminatory patterns [12].
- The trained model can be deployed using Flask or FastAPI.
- A web or mobile interface can be developed for real-time loan prediction.
- The system can be connected to a database for storing application records. Continuous model monitoring can be introduced to detect changes in applicant behavior and data distribution.

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